

Assignment 2: ReAct Agent Report

CE8014 AI 代理系統之設計與開發

2026-03-15

Section 1: Implementation Logic

1.1 Architecture Overview

本專案實作一個 text-based ReAct Agent，透過三階段流程回答複雜問題：

1. **ReAct Loop** - Thought -> Action -> Observation 迴圈，最多 5 步迭代
2. **Evidence Review Gate** - 獨立 LLM 呼叫，驗證 draft answer 是否有充分證據支持
3. **Finalize Stage** - 獨立 LLM 呼叫，進行 entity disambiguation 與最終答案合成

系統架構分為三個檔案：

- **tools.py** - 工具層，提供 search 與 calculate 兩個工具
- **agent.py** - 三階段 ReAct 核心，包含三組 System Prompt、Review Gate、Finalize Stage
- **main.py** - CLI 互動層，使用 prompt_toolkit 提供 slash commands

1.2 System Prompt Strategy

Agent 使用三組獨立的 System Prompt，各司其職：

SYSTEM_PROMPT (ReAct Loop)

主要 prompt，控制 Thought -> Action -> Observation 迴圈行為。設計策略：

1. **工具定義**：明確列出可用工具及其呼叫格式(search[query]、calculate[expression])，讓 LLM 知道自己的能力的邊界。
2. **格式約束**：嚴格定義每步輸出只能是「Thought + Action + PAUSE」或「Answer」二擇一，避免 LLM 自由發揮導致 parse 失敗。
3. **Few-Shot Example**：嵌入一個完整的 ReAct 範例（日本與法國人口比例），展示：
 - 問題拆解 (decomposition)：先搜日本人口，再搜法國人口
 - 工具串接：搜尋結果送入 calculate 計算比例
 - Observation 格式：包含 Domain、URL、Snippet 結構化欄位，與 search summary 標記為 unverified hint
 - 格式遵守：每個 Action 後接 PAUSE，最後用 Answer 結尾

Few-Shot 範例的有效性在於：它與 Task 1（日本 vs 台灣人口）結構高度相似，LLM 能直接遷移此 decomposition pattern。同時，範例中的 Observation 格式與實際搜尋結果格式一致，減少格式適應成本。

4. **Source-First Observation**：搜尋結果格式經過設計，每筆結果包含 Domain / URL / Snippet 三欄，讓 LLM 能根據來源可信度判斷資訊品質。Tavily 的 AI summary 被標記為 unverified hint，引導 LLM 優先信任原始 snippet。

5. **Epistemic Guardrails**：System Prompt 中嵌入多條認知規範：

- **Grounded Assertion** : 「Only assert a fact if you can point to a specific snippet」-防止模型空口宣稱事實
- **Targeted Verification** : 允許使用已找到的線索（人名、URL）進行驗證式搜尋，屬合理調查
- **Anti-Confirmation Bias** : 不假設候選答案正確，需確認來源是否指向問題所述的同一實體
- **Entity Disambiguation** : 「list the distinct entities you see before continuing」-強制模型在同名實體間做區分
- **Inconclusive Fallback** : 允許模型在證據不足時輸出「evidence is inconclusive」，而非強迫猜測

REVIEW_SYSTEM_PROMPT (Evidence Review Gate)

獨立的 LLM 呼叫，接收 question、draft answer、ReAct trace，輸出結構化 JSON：

```
{
  "decision": "accept | revise",
  "reason": "...",
  "next_query": "...",
  "selected_person": "...",
  "supporting_quotes": [...],
  "descriptor_match": true | false,
  "ambiguity_detected": true | false
}
```

核心規則：

- descriptor_match 只有在 evidence 明確匹配問題中的 product/domain 描述時才為 true
- ambiguity_detected 在多個同名實體出現時為 true
- decision 必須在以下任一條件成立時為 revise：descriptor_match 為 false、ambiguity_detected 為 true、selected_person 不被 supporting_quotes 直接支持
- next_query 在 decision 為 revise 時必須非空，應直接針對證據缺口提出具體查詢

FINALIZE_SYSTEM_PROMPT (Final Synthesis)

最終合成階段，進行 entity disambiguation 與答案生成，輸出 JSON：

```
{
  "answer_type": "resolved | inconclusive",
  "selected_person": "...",
  "selected_entity": "...",
  "supporting_quotes": [...],
  "conflicting_entities": [...],
  "final_answer": "..."
}
```

核心規則：

- 不得 invent 不在 observations 中的事實
- supporting_quotes 必須從 observation snippets 中逐字引用
- 若 selected_person 沒有被 supporting_quotes 直接支持，answer_type 必須為 inconclusive
- **Entity Match 驗證**：若問題指定了特定 product type 或 industry（如「AI search」），確認 observations 中的實體確實在該領域營運；若為同名不同實體，設 answer_type 為 inconclusive 並列出 conflicting entities

完整 SYSTEM_PROMPT 如下：

You are a research assistant that answers questions using a Thought -> Action -> Observation loop (ReAct pattern).

You have access to the following tools:

- search[query]: Search the web for current information. Use specific, targeted queries.
- calculate[expression]: Evaluate a math expression (supports +, -, *, /, **).

On each step you MUST output exactly ONE of:

1. A Thought line followed by an Action line, then PAUSE (wait for observation)
2. An Answer line (when you have enough information)

Format rules:

- Thought: <your reasoning>
- Action: <toolname>[<argument>]
- PAUSE
- Answer: <your final answer>

IMPORTANT:

- When you have enough information to answer, you MUST start your response with "Answer:" on its own line. Do NOT output your conclusion without the "Answer:" prefix -- the system cannot parse it otherwise.
- After writing an Action line, you MUST write PAUSE and stop. Do NOT write Observation yourself.
- The system will provide the Observation after your PAUSE.
- Treat search summaries as hints, not as verified facts. Prefer source titles, domains, and snippets.
- Only assert a fact if you can point to a specific snippet from the Observations that states it. If no snippet explicitly supports a claim, treat it as unverified.
- If a search returns poor, conflicting, or off-topic results, reflect on why and try a DIFFERENT query angle.
- When different search results appear to refer to different

- entities with the same name, list the distinct entities and refine your query with neutral qualifiers (industry, product type, headquarters, or domain) before answering.
- When following up on a lead (e.g., a person's name or URL found in results), you MAY use it in a targeted verification query. This is legitimate investigation, not confirmation bias.
Example: `search["morphic.sh" "Yoshiki Miura"]`
 - AVOID CONFIRMATION BIAS: Do not assume a candidate answer is correct just because one source mentions it. Always check whether the source refers to the SAME entity the question asks about.
 - If ambiguity remains after a reasonable retry, give an Answer that says the evidence is inconclusive and briefly list the conflicting entities you found, instead of guessing.
 - Break complex questions into sub-questions and solve them step by step.

Example session:

Question: What is the population ratio of Japan to France?

Thought: I need to find the populations of both Japan and France. Let me start with Japan.

Action: `search[Japan population 2024]`

PAUSE

Observation:

[1] RESULT

Title: Japan Demographics

Domain: `example.com`

URL: `https://example.com/japan`

Snippet: Japan's population was estimated at 123.3 million in 2024, continuing a declining trend.

Search summary (unverified hint; prefer the source snippets above if they conflict): Japan's population is approximately 123.3 million as of 2024.

Thought: Japan's population is about 123.3 million. Now I need France's population.

Action: `search[France population 2024]`

PAUSE

[...observation, calculate, final answer...]

Answer: The population ratio of Japan to France is

approximately 1.80:1.

Python 如何將 **Observation** 回饋至 **LLM context window**

ReAct Loop 以 messages list 維護完整對話歷史。每次工具呼叫後，Observation 作為 user role 訊息注入：

```
# LLM 輸出的 Action 記為 assistant message
messages.append({"role": "assistant", "content": raw + "\nPAUSE"})

# 工具執行結果作為 user message 注入
tool_func = TOOL_REGISTRY[tool_name]
observation = tool_func(tool_arg)
messages.append({"role": "user", "content": f"Observation: {observation}"})
```

這確保 LLM 在每個 iteration 都能看到完整的 Thought-Action-Observation 歷史。Review Gate 和 Finalize Stage 各自使用獨立的 messages（只包含自己的 system prompt + 從主 loop 提取的 trace），不會污染主 loop 的 context。

1.3 Loop Mechanism

三階段流程

```
messages = [system_prompt, user_query]
draft_answer = None
review_budget = 2 # review gate 最多觸發兩次 retry search

for i in 1..MAX_ITERATIONS(5):
    response = LLM(messages, stop=["PAUSE", "Observation:"])

    if "Answer:" in response:
        draft_answer = extract_answer(response)

        # --- Stage 2: Evidence Review Gate ---
        review = _review_draft(question, messages, draft_answer)

        if review.decision == "accept":
            break # proceed to finalize

        if i < MAX and review_budget > 0:
            review_budget -= 1
            run_search(review.next_query) # one targeted retry
            continue

    break # budget exhausted, go to finalize anyway
```

```

    if "Action:" in response:
        result = dispatch_tool(action, arg)
        messages.append(observation)
        continue

    messages.append(nudge_message)

# --- Stage 3: Finalize ---
final_answer = _finalize_answer(question, messages, draft_answer)
return final_answer

```

Stage 2: Evidence Review Gate (_review_draft)

Review Gate 分為兩層檢查：

第一層：Structural Guard (結構化前置檢查) - 在呼叫 LLM 之前，Python 端計算目前 messages 中的 observation 數量。若 obs_count < 2，直接返回 revise，跳過 LLM review。這是防止模型在只有一筆搜尋結果時就過早收斂的安全網。當 structural guard 觸發時，follow-up query 由獨立 LLM 呼叫 (_suggest_followup_query) 生成，使用中性 descriptor 避免 confirmation bias。

第二層：LLM 語意審查 - 當 observation >= 2 時，進入完整的 LLM review。使用 REVIEW_SYSTEM_PROMPT，接收 question、draft answer、ReAct trace，輸出結構化 JSON。Python 端只做基本結構驗證 (確認 decision 為 accept 或 revise)，語意判斷委託給 LLM。

review_budget = 2 限制 review gate 最多觸發兩次 retry search，為 entity disambiguation 提供足夠的搜尋空間。若 review 判定 accept，agent 直接回傳 draft answer，跳過 finalize 階段以避免不必要的 API 呼叫和答案漂移。

Stage 3: Finalize (_finalize_answer)

Finalize 同樣是獨立 LLM 呼叫，使用 FINALIZE_SYSTEM_PROMPT。Python 端信任 LLM 的結構化輸出：若 answer_type 為 inconclusive 則使用 inconclusive 答案 (含 conflicting_entities 資訊)，否則直接採用 final_answer。

Stop Sequence

API 呼叫時設定 stop=["PAUSE", "Observation:"]，這是防止 LLM hallucinate Observation 的關鍵機制：

- **PAUSE**：主停止點。LLM 寫完 Action 後應輸出 PAUSE，API 在此截斷回應。
- **Observation**：Fallback 停止點。若 LLM 試圖自行編造 Observation，API 會在此截斷。

被截斷後，程式端補回 PAUSE 到 messages history，確保 LLM 在後續 iteration 看到一致的格式。

Parse Logic

使用 regex 解析 LLM 輸出：

- Answer: (.+) -> 進入 Review Gate
- Action: (\w+)\[(.+?)\] -> 呼叫對應工具，將結果作為 Observation 注入
- 都沒匹配 -> 送 nudge 訊息提醒 LLM 使用正確格式，不直接中斷

Nudge Mechanism

當偵測到「既無 Action 也無 Answer」時，不中斷迴圈，而是注入提示訊息讓 LLM 在下一個 iteration 用正確格式補上回答。

1.4 Tool Design

search

使用 Tavily Python SDK，設定 `search_depth="advanced"`、`max_results=5`、`include_answer=True`。回傳格式為 `source-first` 結構：

```
[1] Title
    Domain: example.com
    URL: https://...
    Snippet: ...
```

Search summary (unverified hint; prefer the source snippets above if they conflict): ...

每筆結果包含 Domain / URL / Snippet，讓 LLM 能根據來源判斷可信度。Tavily AI summary 標記為 `unverified hint`，避免模型過度信任合成摘要。

Snippet 長度上限 500 字元，超過部分截斷並加 `...`，防止單一搜尋結果佔用過多 context window。錯誤一律轉為字串回傳，讓 Agent 自行 reflect 而非 crash。

calculate

使用 Python `ast` 模組安全解析數學運算式，只允許 `+`、`-`、`*`、`/`、`**` 五種運算符。透過遞迴走訪 AST 節點實現，完全阻擋 `eval injection`（如 `__import__("os")`）。

Section 2: Benchmark Traces

以下為三個 Task 的完整 console trace 與分析。所有 Task 使用同一個 ReActAgent 類別，每次 `run()` 呼叫重置 messages（三個 Task 為獨立問題，不需跨問題記憶）。

Task 1: Population Fraction

Question: *What fraction of Japan's population is Taiwan's population as of 2025?*

Console Trace

```
--- Step 1/5 ---
Thought: I need to find the populations of both Japan and Taiwan for the year 2025. I'll start with Japan's population.
Action: search[Japan population 2025]
Observation:
[1] Japan Population 2025 – UNFPA
    Domain: unfpa.org
    Snippet: Total Population (2025): 123,100,000
[2] Japan Population, 1899–2025
    Domain: ceicdata.com
... (19 more lines)

--- Step 2/5 ---
Thought: Japan's population in 2025 is approximately 123 million. Now, I need to find Taiwan's population for the same year.
Action: search[Taiwan population 2025]
Observation:
[1] Taiwan Population, 1950–2025 – CEIC Data
    Domain: ceicdata.com
[2] Taiwan Population – Trading Economics
    Domain: tradingeconomics.com
... (19 more lines)

--- Step 3/5 ---
Thought: Japan's population is approximately 123 million, and Taiwan's population is about 23.3 million.
Action: calculate[23.3 / 123]
Observation: 0.18943089430894308

--- Step 4/5 ---
Thought: Draft answer passed evidence review.

Answer: The fraction of Japan's population that is Taiwan's population as of 2025 is approximately 0.189, or about
```

18.9%.

Analysis

- **Decomposition** : Agent 正確將問題拆解為三個子任務-搜尋日本人口、搜尋台灣人口、計算比例。
- **Tool Chaining** : 搜尋結果成功傳遞給 `calculate` 工具，展現跨工具協作能力。
- **Few-Shot** 效果 : Task 1 的結構與 System Prompt 中的 Few-Shot 範例 (日本 vs 法國) 高度相似，LLM 完美複製了這個 `decomposition pattern`。
- **Review Gate** : draft answer 直接通過 review (observation ≥ 2 、無歧義、data grounded)。
- 步數 : 4 步完成 (2 次搜尋 + 1 次計算 + 1 次 answer)，效率合理。

Task 2: Display Specs Comparison

Question: Compare the main display specs of iPhone 15 and Samsung S24.

Console Trace

```
--- Step 1/5 ---
Thought: I need to find the display specifications for both
the iPhone 15 and the Samsung S24. I'll start with iPhone 15.
Action: search[iPhone 15 display specifications]
Observation:
  [1] iPhone 15 Pro Max – Tech Specs
      Domain: support.apple.com
      Snippet: Super Retina XDR OLED 6.7" 2796x1290 460ppi
      ProMotion 120Hz, HDR, 2000 nits peak brightness
  [2] iPhone 15 – Wikipedia
      Domain: en.wikipedia.org
  ... (19 more lines)

--- Step 2/5 ---
Thought: I have the display specifications for the iPhone 15.
Now, I need to find the display specs for Samsung S24.
Action: search[Samsung S24 display specifications]
Observation:
  [1] Galaxy S24 | Samsung Mobile Press
      Domain: samsungmobilepress.com
      Snippet: 6.2" FHD+ Dynamic AMOLED 2X, Super Smooth
      120Hz refresh rate (1~120Hz), Vision booster
  [2] Samsung Galaxy S24 – GSMArena
      Domain: gsmarena.com
  ... (19 more lines)

--- Step 3/5 ---
Thought: I have gathered the display specifications for both
devices. Now, I can summarize and compare their main specs.
[!] No Action or Answer detected, nudging LLM...

--- Step 4/5 ---
Thought: Draft answer passed evidence review.

Answer:
iPhone 15: Super Retina XDR OLED, 6.1", 2556x1179, 60Hz,
2000 nits, HDR, True Tone, Dynamic Island
Samsung S24: Dynamic AMOLED 2X, 6.2", 1080x2340, 120Hz,
2600 nits, HDR10+, Vision Booster, Gorilla Glass Victus 2

The iPhone 15 has a slightly smaller display with a higher
resolution and lower refresh rate, while the Samsung S24 has
```

a higher refresh rate and peak brightness.

Analysis

- **Search Strategy** : Agent 分別搜尋兩支手機的規格，共兩次搜尋取得完整資訊。
- **具體規格差異** : 成功擷取到 60Hz vs 120Hz refresh rate、resolution 差異 (2556x1179 vs 1080x2340)、亮度差異 (2000 vs 2600 nits)、HDR 支援差異等關鍵規格。
- **格式遵從問題** : Step 3 中 LLM 直接輸出比較內容但遺漏 **Answer** : 前綴，觸發 nudge 機制。這是 gpt-4o-mini 在 instruction following 上的常見問題-當資訊充足時，模型傾向直接輸出答案而忽略格式約束。Nudge 機制在此展現了其必要性。
- **步數** : 4 步完成 (2 次搜尋 + 1 次 nudge + 1 次 answer)。

Task 3: Morphic CEO (Resilience & Reflection Test)

Question: *Who is the CEO of the startup 'Morphic' AI search?*

此 Task 是最具挑戰性的測試，因為「Morphic」是多個不同公司或專案的名稱：

- **morphic.sh** – 開源 AI search engine (問題所指的目標)，由 Yoshiki Miura 開發維護 (GitHub: miurla)，屬個人開源專案
- **morphic.com** – AI 影片/動畫平台 (Jaynti Kanani, Polygon 前共同創辦人)
- **getmorphic.com** – AI 網站建置平台 (Weston Baker, Founder)
- **Morphic Therapeutic** – 生技公司 (Praveen Tipirneni, CEO)
- **Morphic Medical** – 醫療器材公司 (Mike Gutteridge, CEO)

搜尋引擎對「Morphic CEO」的回傳結果會混合這些不同實體，構成 entity disambiguation 挑戰。

Trace A: gpt-4o-mini (Incorrect –Entity Mismatch)

```
--- Step 1/5 ---
Thought: I need to find the current CEO of the startup
'Morphic' that focuses on AI search.
Action: search["Morphic AI search CEO"]
Observation:
[1] Morphic – 2026 Company Profile & Team – Tracxn
    Domain: tracxn.com
    URL: tracxn.com/d/companies/morphic/...
    Snippet: Jaynti Kanani is the current CEO of Morphic.
    Headquartered in San Francisco, United States.
[2] Morphic Management Team – CB Insights
    Domain: cbinsights.com
... (19 more lines)

--- Step 2/5 ---
Thought: There seems to be conflicting information. One
source states Jaynti Kanani, while another claims Weston
Baker. I need to verify.
Action: search["Morphic AI Weston Baker Jaynti Kanani
CEO confirmation"]
Observation:
[1] Polygon Cofounder Jaynti Kanani Steps Down...
    Domain: inc42.com
    URL: inc42.com/buzz/polygon-cofounder-jaynti-kanani...
[2] Morphic Launches AI Anime 'DQN' & $1M Fund
    Domain: animationmagazine.net
... (19 more lines)

--- Step 3/5 ---
Thought: The search results still show conflicting info.
I need to clarify the current leadership structure.
```

Action: search["Morphic AI leadership team 2026"]

Observation:

- [1] Morphic – 2026 Company Profile & Team – Tracxn
Domain: tracxn.com
Snippet: Jaynti Kanani is the current CEO of Morphic.
- [2] Meet The Morphic Management Team
Domain: morphicmedical.com
- ... (19 more lines)

--- Step 4/5 ---

Thought: Jaynti Kanani is indicated as CEO of Morphic.
Let me confirm he is the CEO of Morphic AI, not Medical.

Action: search["Morphic AI Jaynti Kanani CEO
confirmation 2026"]

Observation:

- [1] Morphic to Produce AI Anime Series... – Variety
Domain: variety.com
URL: variety.com/2025/global/news/morphic-ai-anime...
Snippet: ...said Jaynti Kanani, former CEO and founder of blockchain company Polygon, who launched Morphic in 2024 with bases in San Jose, New York, Dubai...
- [2] Morphic Launches AI Anime 'DQN' & \$1M Fund
Domain: animationmagazine.net
- ... (19 more lines)

--- Step 5/5 ---

[!] Evidence review found gaps; review budget exhausted.
Moving to final synthesis.

--- Finalize ---

Thought: Synthesizing evidence and resolving ambiguity.

Answer: The current CEO of Morphic, a startup focused on AI search, is Jaynti Kanani. He has been leading the company since its launch in 2024, and Morphic is dedicated to empowering creators and filmmakers with advanced technology.

Trace B: MiniMax M2.5 (Inconclusive –正確辨識歧義)

為對照不同模型在相同架構上的表現，以相同程式碼切換至 MiniMax M2.5 進行測試：

--- Step 1/5 ---

Thought: I need to find information about a startup called "Morphic" that is related to AI search, and who its CEO is.

Action: search[Morphic AI search startup CEO]

Observation:

```
[1] Morphic – 2026 Company Profile & Team – Tracxn
    Domain: tracxn.com
    URL: tracxn.com/d/companies/morphic/...
    Snippet: Jaynti Kanani is the current CEO of Morphic.
[2] Eric Vardon – Morphio.ai (different company)
    Domain: youtube.com
... (19 more lines)
```

--- Step 2/5 ---

Thought: The search results show conflicting information and different companies named "Morphic". Let me search more specifically for the AI search engine startup.

Action: search["Morphic" AI search engine startup]

Observation:

```
[1] Introducing Morphic: AI-Powered Search Engine
    Domain: linkedin.com
    URL: linkedin.com/posts/hamza-abar...
[2] AI startup Morphic unveils AI-powered anime DQN...
    Domain: instagram.com
... (19 more lines)
```

--- Step 3/5 ---

Thought: The results show Morphic is related to AI anime production, not necessarily AI search. But the first result mentions "Morphic: AI-Powered Search Engine".

Action: search[Morphic AI search engine founder CEO]

Observation:

```
[1] Weston Baker – Morphic
    Domain: linkedin.com
    URL: linkedin.com/in/westonbaker
    Snippet: Founder @ Morphic | AI-Powered Websites
    for VCs & Growth Firms
[2] Morphic Launches AI Anime 'DQN' & $1M Fund
    Domain: animationmagazine.net
... (19 more lines)
```

--- Step 4/5 ---

Thought: There are multiple companies named "Morphic" with different focuses. I need to find specifically the "Morphic" that is an AI search startup.

Action: search["Morphic" "AI search" startup company]

Observation:

```
[1] AI startup Morphic unveils AI-powered anime DQN...
    Domain: notebookcheck.net
    URL: notebookcheck.net/.../morphic-ai-anime...
[2] Morphic | LinkedIn
    Domain: linkedin.com
... (19 more lines)
```

```
--- Step 5/5 ---
Thought: I found morphic.sh -- an "open-source AI search
engine." Let me get more info about its CEO.
Action: search[morphic.sh founder CEO]
Observation:
  [1] Founder Stories: Praveen Tipirneni, CEO of
      Morphic Therapeutic
      Domain: youtube.com
  [2] Who is the CEO of Morphic Therapeutic?
      Domain: clay.com
  ... (19 more lines)

--- Finalize ---
Thought: Synthesizing evidence and resolving ambiguity.
Finalize: answer_type=inconclusive

Final: The observations show multiple companies named
"Morphic" operating in different domains:
- morphic.sh (AI search engine, open-source)
- morphic.com (Jaynti Kanani, AI anime production)
- getmorphic.com (Weston Baker, AI web design)
- Morphic Therapeutic (Praveen Tipirneni, biotech)
Without a verbatim quote from morphic.sh specifically
identifying their leadership, the answer remains
inconclusive.
```

Task 3 Analysis

gpt-4o-mini 的行為：

- Step 1-2 正確識別了搜尋結果中的衝突 (Jaynti Kanani vs Weston Baker)，展現初步 reflection
- Step 2 使用 targeted verification query ("Morphic AI Weston Baker Jaynti Kanani CEO confirmation")，將兩個候選人名嵌入搜尋。這是新版 prompt 允許的合法調查行為，但模型未能從結果中區分不同實體
- Step 3-4 持續搜尋，Step 4 找到 Variety 文章明確描述 Jaynti Kanani 為 Polygon 前共同創辦人、2024 年成立 Morphic 做 AI anime –但模型忽略了「anime」這個關鍵 product type descriptor
- **Entity Mismatch** 未被偵測：模型錯誤地認定 Jaynti Kanani 為「Morphic AI search」的 CEO，未區分 morphic.com (AI 影片/動畫平台) 與 morphic.sh (AI 搜尋引擎)
- Review Gate 在 Step 5 偵測到證據缺口，但 review budget 已用完，進入 Finalize
- **Finalize** 未能修正錯誤：Finalize 階段也使用 gpt-4o-mini，同樣未能辨識 entity mismatch，最終輸出錯誤答案
- 與舊版 prompt 相比：舊版收斂在 Weston Baker (getmorphic.com 網頁設計)，新版收斂在 Jaynti Kanani (morphic.com AI anime) –錯誤實體不同，但核心問題相同

MiniMax M2.5 的行為：

- 第一步即識別出搜尋結果包含不同公司，主動縮窄搜尋範圍
- Step 2 使用中性 query ("Morphic" AI search engine startup) 找到 morphic.sh 相關的 LinkedIn 文章
- Step 3 找到 Weston Baker (getmorphic.com, AI 網站建置)，正確辨識其為不同 Morphic
- Step 4-5 逐步縮窄至 morphic.sh，但搜尋結果被 Morphic Therapeutic (生技公司) 干擾
- Finalize 正確輸出 inconclusive，列出四個不同 Morphic 實體及其各自領域

Reflection 路徑對比：

階段	gpt-4o-mini	MiniMax M2.5
Entity 識別	識別衝突但未深入區分	逐步區分不同實體
搜尋策略	4 次搜尋，嵌入候選人名驗證	5 次搜尋，逐步縮窄至 morphic.sh
Targeted Verification	使用但導致錯誤收斂	未嵌入候選人名，穩定使用中性 query
Entity Mismatch	未偵測 (anime vs AI search)	正確區分不同 product type
Review Gate	Step 5 偵測到 gap，budget 用完	未進入 (5 步用完)
最終結果	Incorrect (Jaynti Kanani)	Inconclusive (正確的保守輸出)

Section 3: Model Capability Analysis

跨模型對照實驗

使用相同 agent 程式碼（同一個 ReActAgent class、同一組 system prompt），僅透過 .env 切換 MODEL 環境變數，對三個 Task 進行跨模型測試：

Model	Task 1	Task 2	Task 3
gpt-4o-mini	正確 (18.9%)	正確 (需 nudge)	錯誤 (entity mismatch)
MiniMax M2.5	正確 (18.9%)	正確 (一次通過)	Inconclusive (正確消歧)

發現

1. 格式遵守能力差異：

gpt-4o-mini 在 Task 2 (Step 3) 中觸發了 nudge - 模型在資訊充足時傾向直接輸出比較結果而忽略 Answer: 前綴。Task 3 中因 SYSTEM_PROMPT 加入了「you MUST start your response with Answer:」的強制指示，nudge 未再發生。MiniMax M2.5 在相同架構下從未出現此問題。

2. Entity Disambiguation 能力差距：

Task 3 是區分模型推理能力的關鍵測試。核心差異在於 **product type mismatch** 偵測：

gpt-4o-mini：搜尋到 Variety 文章明確描述 Jaynti Kanani「launched Morpnic in 2024」做 AI anime 後，直接認定其為「Morpnic AI search」的 CEO。模型未能區分「AI anime production」(morpnic.com) 與「AI search engine」(morpnic.sh)。即使使用了 targeted verification query（嵌入候選人名搜尋），模型在 4 次搜尋後仍未能正確區分 product type - 這是語意推理層面的能力不足，而非搜尋策略問題。

MiniMax M2.5：在 Step 3 中明確區分了 Weston Baker (getmorpnic.com 網頁設計)，Step 4-5 逐步縮窄至 morpnic.sh 但搜尋結果被其他同名實體干擾，在無法確認 morpnic.sh 具體開發者的情況下正確輸出 inconclusive。

3. Structural Guard 的效果與局限：

_review_draft 中的 observation 數量門檻 (obs_count < 2) 有效防止了 gpt-4o-mini 在第一次搜尋後就過早收斂。但結構化檢查只能確保「搜了足夠多次」，無法確保「搜到正確的東西」- 當模型在語意層面無法區分同名不同產品的實體時，再多的搜尋也不會改善結果。這驗證了 structural guard 是必要的安全網，但不是萬能的解決方案。

4. Inconclusive 是合理的安全輸出：

MiniMax M2.5 在 Task 3 的結果是 inconclusive。morpnic.sh 實際上是 Yoshiki Miura 的個人開源專案，搜尋結果中出現的 Jaynti Kanani (morpnic.com, AI 影片平台) 和 Weston Baker (getmorpnic.com, 網站建置) 都是不同的 Morpnic。在此條件下輸出 inconclusive 並列出衝突實體，是合理的判斷。

5. 架構設計原則-分層防禦：

Python orchestration 層設置最小限度的結構化檢查（如 observation 數量門檻）作為安全網，核心語意判斷委託 LLM structured output 處理。過多的 Python-side 語意驗證容易產生 false negative，過少則無法防止模型過早收斂。

Summary

	Task 1	Task 2	Task 3 (4o-mini)	Task 3 (MiniMax)
Steps Used	4/5	4/5	5/5	5/5
Search Calls	2	2	4	5
Calculate Calls	1	0	0	0
Decomposition	Yes	Yes	N/A	N/A
Reflection	No (unnecessary)	No (unnecessary)	Yes (partial)	Yes (full)
Nudge Triggered	No	Yes (Step 3)	No	No
Review Gate	Accept	Accept	Revise (too late)	N/A (5 步用完)
Result	Correct	Correct	Incorrect	Inconclusive

Task 1、2 在 4 步內主動完成；Task 3 兩個模型都用滿 5 步上限，是被迫結束而非主動收斂—這反映了 entity disambiguation 問題的搜尋複雜度超出 5 步預算。整體而言，三個 Task 展現了 ReAct pattern 的核心能力與局限：

1. **Task Decomposition**：複雜問題（Task 1, 2）被自動拆解為多個子問題依序處理。
2. **Tool Selection**：Agent 根據子問題性質選擇適當工具（搜尋 vs 計算）。
3. **Resilience & Reflection**：Task 3 展示了 self-correction 流程的完整光譜—gpt-4o-mini 在 Thought 中識別了搜尋結果的衝突（reflection），並使用 targeted verification query 嵌入候選人名搜尋，但在語意推理層面未能區分 AI anime (morphic.com) 與 AI search (morphic.sh)，最終給出錯誤答案；MiniMax M2.5 則透過更精確的 entity disambiguation 和中性搜尋策略，正確判定證據不足並輸出 inconclusive。
4. **Model-Agnostic Architecture**：同一套 agent 程式碼在不同模型上展現不同水平的表現，驗證了架構設計的可遷移性。Structural guard 提供了最低安全保障，但最終結果品質仍取決於模型本身的語意推理能力。